1.Title of the Project

Development of a feature-rich, practical online medicine shop (PHARMA).

2. Introduction and Objectives of the Project

In the digital age, the healthcare industry is rapidly evolving, and online medicine shops are becoming increasingly popular. These platforms provide a convenient and efficient way for customers to purchase medicines and healthcare products from the comfort of their homes. An online medicine shop allows users to browse through a wide range of products, place orders, and have them delivered directly to their doorsteps, often with just a few clicks. This system also provides valuable information about medicines, their usage, and availability, ensuring that customers make informed decisions.

The project aims to develop an online medicine shop that caters to the needs of customers by offering an intuitive user interface, a secure payment gateway, and a comprehensive product catalog. The platform will be designed to handle various functionalities such as product search, category filtering, user authentication, order processing, and inventory management.

1. **A person should be able to**

    - **Access information from the shop page (only read access).**
    - **login to the system through the first page of the application**
    - **See/change his/her details (like how many reports/technical documents/submitted)**
    - **get help about Pharma on how to use the different features of the system**

2. **An admin login should be present who can read, approve as well as remove any uploads.**

3. **Project Category**

Web Application

4. **Tools/Platform, Hardware and Software Requirement specifications.**

Tools

1. VS Code, Browser

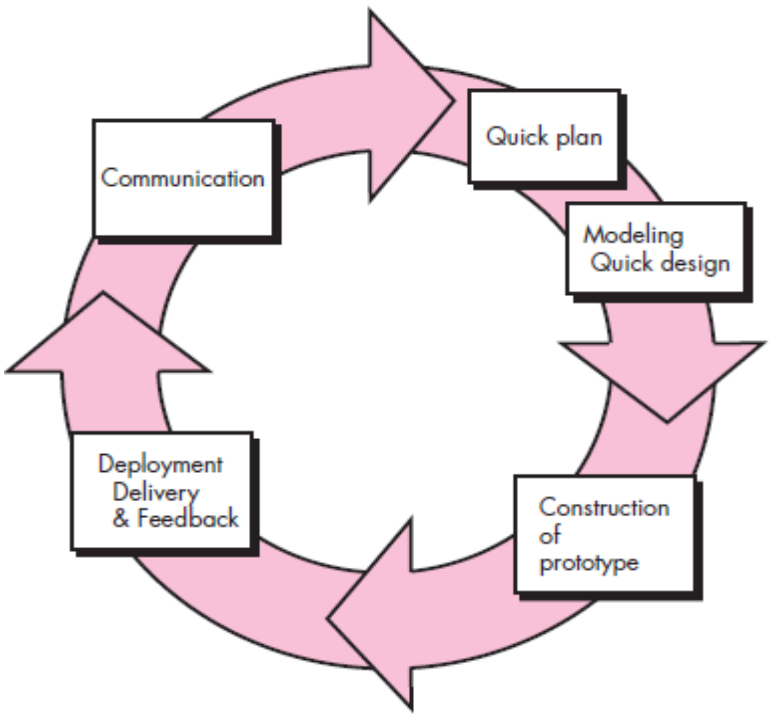
Platform

1. Microsoft Windows 10/11

5. Goals of Implementation

The implementation aims at seamless document sharing across the institution.

1. SDLC Process Applied

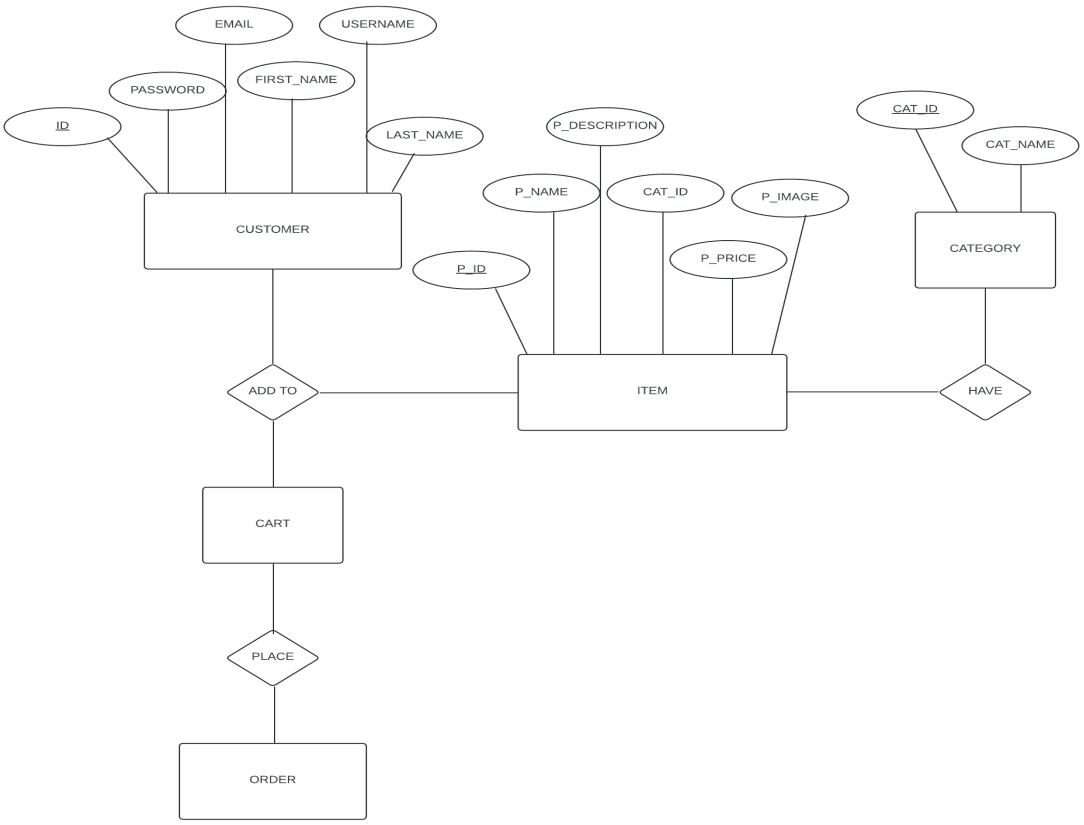


Often, a customer defines a set of general objectives for software but does not identify detailed input, processing, or output requirements. In other cases, the developer may be unsure of the efficiency of an algorithm, the adaptability of an operating system, or the form that human/machine interaction should take. In these, and many other situations, a prototyping paradigm may offer the best approach.

The prototyping paradigm begins with requirements gathering. Developer and customer meet and define the overall objectives for the software, identify whatever requirements are known, and outline areas where further definition is mandatory. A "quick design" then occurs. The quick design focuses on a representation of those aspects of the software that will be visible to the customer/user (e.g., input approaches and output formats). The quick design leads to the construction of a prototype. The prototype is evaluated by the customer/user and used to refine requirements for the software to be developed. Iteration occurs as the prototype is tuned to satisfy the needs of the customer, while at the same time enabling the developer to better understand what needs to be done.
Ideally, the prototype serves as a mechanism for identifying software requirements. If a working prototype is built, the developer attempts to use existing program fragments or applies tools (e.g., report generators, window managers) that enable working programs to be generated quickly.

2. **Data Model**

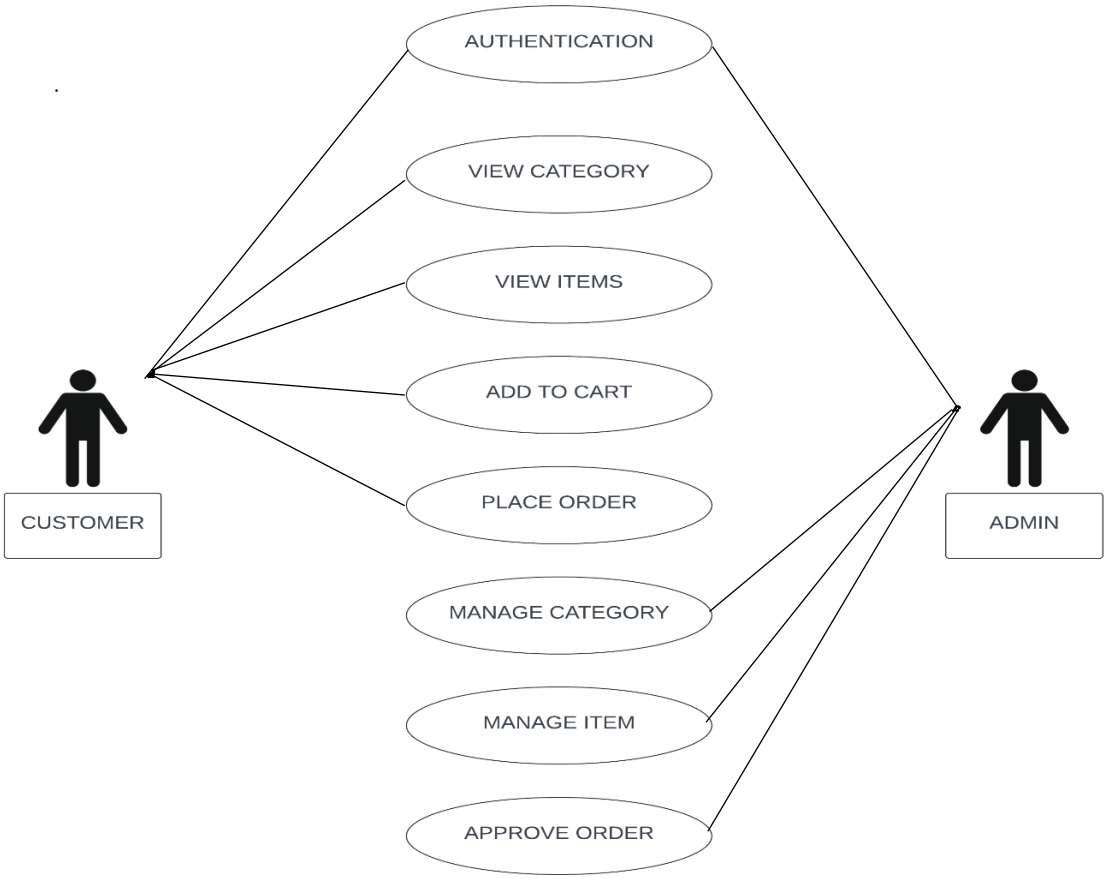
ER DIAGRAM



3. **Functional Requirements**

The online medicine shop must support secure user authentication, enabling users to register, log in, and manage their accounts. It should include a robust product management system that allows for the addition, updating, and removal of products, with details such as images, prices, and descriptions. The platform must efficiently handle order processing, from payment and confirmation to delivery tracking. Additionally, the website should offer filtering features to help users quickly locate products based on categories and prices.

4. Use Case Diagram



5. Use Case Descriptions

1. **AUTHENTICATION**: Allows users (both Manager and Customer) to authenticate and gain access to the system also enables users to exit the system securely and allows new customers to create an account in the system.

2. **VIEW CATEGORY**: Allow users to view the categories of the items. User can also sort the items by the category.

3. **VIEW ITEMS**: Allow users to see items.

4. **ADD TO CART**: Customers can add selected products to their shopping cart before purchasing.

5. **PLACE ORDER**: Allows users to order products from the pharmacy.

6. **MANAGE CATEGORY**: Admin can manage (Add / Delete) all categories.

7. **MANAGE ITEMS**: Admin can manage (Add / Delete / Update) items.

8. **APPROVE ORDER**: Admin approve the orders of the users.

6. **Non Functional Requirements**

1. Performance

- **Response Time**: When a customer searches for medicine on PHARMA, the search results should appear almost instantaneously, with no more than a blink of an eye (2 seconds) delay. Even during high-traffic times, like a morning rush when customers order their daily prescriptions, the website should feel just as responsive as when it's quiet.
- **Scalability**: Imagine PHARMA's annual health sale, where thousands of customers flock to the site simultaneously. Despite the surge, the system remains smooth and fast, handling the influx without breaking a sweat, seamlessly scaling to accommodate up to 10,000 customers at once.

2. Security

- **Data Protection**: Visualize every piece of sensitive customer data (like credit card details and health information) being wrapped in a digital vault, shielded by advanced encryption that makes it impervious to cyber-attacks, ensuring customers' peace of mind.
- **User Authentication**: Picture logging into PHARMA: after entering your password, a quick confirmation on your mobile device adds an extra layer of security, making sure that no one but you can access your account.
- **Access Control**: Think of PHARMA as a digital fortress where managers and customers enter through different gates. Managers can manage the entire system from behind their secure gate, while customers have their own secure access, only able to see and do what is relevant to their shopping experience.

3. Availability

- **Uptime**: Imagine PHARMA as a 24/7 virtual pharmacy that never sleeps, where the lights are always on, ready to serve customers at any hour, with downtime being almost as rare as a lunar eclipse—just 0.1% of the year.

- **Backup**: Picture a nightly ritual where PHARMA quietly duplicates all its critical data into a secure, off-site vault, ensuring that even in the worst-case scenario, every piece of information is safe and can be restored effortlessly.

4. Usability

- **Ease of Use**: Envision a customer visiting PHARMA for the first time, yet feeling like they've shopped there a hundred times before. The interface is so intuitive and user-friendly that they effortlessly find, select, and purchase their needed medications within minutes.
- **Accessibility**: Picture PHARMA as a welcoming, all-inclusive environment, where customers with visual impairments easily navigate and order through voice commands, and the website adapts smoothly to various assistive technologies.

5. Maintainability

- **Code Quality**: Imagine PHARMA’s backend as a beautifully organized library where every piece of code is a well-labeled book, making it easy for developers to find and update information without any confusion or errors.
- **Documentation**: Think of PHARMA’s documentation as a comprehensive guidebook, clear and detailed enough that a new developer or customer service agent can understand the system’s inner workings or troubleshooting steps with ease.

6. Reliability

- **Error Handling**: Visualize PHARMA as a resilient system where, if something goes wrong, it doesn’t panic. Instead, it gracefully informs the customer with a friendly message and logs the issue behind the scenes, ready for quick resolution by the technical team.
- **Recovery**: Picture PHARMA as a digital phoenix: in the rare event of a crash or system failure, it rises from the ashes within minutes, with all data intact and business continuing as usual.

7. Scalability

- **Load Handling**: Imagine PHARMA as an elastic band that stretches effortlessly when more customers join, handling peak

times like national health days or pandemic surges with ease, and then contracting smoothly during off-peak hours.
- **Elasticity**: Think of PHARMA as a smart energy saver that scales down its operations during low traffic, ensuring resources are used efficiently, without compromising performance.

8. Compliance

- **Legal Compliance**: Envision PHARMA as a digital fortress, where every transaction and data process strictly follows international laws, like GDPR, ensuring customer data is treated with the highest standard of care and privacy.
- **Industry Standards**: Picture PHARMA's checkout process as a secure, high-standard financial institution where every payment is processed following the stringent PCI DSS guidelines, ensuring that customers' financial data is as secure as possible.

9. Interoperability

- **API Integration**: Imagine PHARMA as a well-connected hub in a bustling digital marketplace, where its APIs allow seamless integration with payment gateways, delivery services, and even health apps, ensuring a smooth and interconnected experience for all users.
- **Cross-Platform Compatibility**: Picture PHARMA being equally efficient and visually appealing whether you're accessing it on a desktop in the office, a tablet in a clinic, or a smartphone in the comfort of your home.

10. Performance Monitoring

- **Real-time Monitoring**: Visualize PHARMA as a high-tech control room, with real-time dashboards monitoring every aspect of the system, ready to alert administrators if something is amiss, allowing them to resolve issues before customers even notice.
- **Analytics**: Picture PHARMA as a data-driven machine, collecting insights on user behaviour, popular products, and website performance, using this data to continuously improve the user experience and optimize business strategies.

7. Feasibility Study

1. Technical Feasibility

- **Technology Stack:** PHARMA is planned to be built using a modern, robust technology stack. The backend will utilize Django, leveraging its powerful admin interface for managing inventory and orders, while the frontend will be crafted using React, ensuring a responsive and dynamic user experience. The technical team is proficient in these technologies, minimizing the learning curve and reducing development time.
- **Infrastructure:** PHARMA will be hosted on AWS, taking advantage of its scalability and reliability. This setup will include load balancers to manage traffic efficiently during peak times, such as during promotional sales or seasonal demand spikes. The infrastructure is designed to scale automatically, ensuring consistent performance as the customer base grows.

2. Economic Feasibility

- **Initial Costs:** The estimated initial investment for PHARMA is $50,000. This includes development costs ($30,000), infrastructure setup ($10,000), and initial marketing efforts ($10,000). Given the rising trend in online healthcare services, the projected return on investment (ROI) is expected within 18 months, with a break-even point anticipated after the first year of operations.

- **Revenue Projections:** PHARMA plans to generate revenue through direct sales of pharmaceutical products, with an expected average order value of $60. Additionally, a subscription model for regular customers, offering discounts on recurring orders, is projected to contribute 20% to the total revenue. The revenue is projected to grow by 15% per quarter, with a significant increase expected during the first flu season after launch.

3. Market Feasibility

- **Market Demand:** The online pharmacy market is growing rapidly, especially in the wake of the global pandemic, which has driven more customers to seek healthcare solutions online. PHARMA is positioned to capitalize on this trend by offering not only medicines but also health supplements and wellness products, catering to a broad demographic.
- **Competitor Analysis:** PHARMA will face competition from established online pharmacies like CVS and Walgreens. However, PHARMA's unique selling point (USP) will be its user-friendly interface, a wide range of products, and a focus on personalized customer service, including 24/7 online pharmacist consultations. By targeting niche markets, such as natural remedies and homeopathic medicines, PHARMA can differentiate itself and build a loyal customer base.

4. Legal Feasibility

- **Regulatory Compliance:** Operating an online pharmacy requires strict adherence to local and international pharmaceutical regulations. PHARMA will obtain all necessary licenses to sell prescription and over-the-counter (OTC) medicines. A legal team has been engaged to ensure compliance with GDPR for European customers and HIPAA regulations for U.S. customers, safeguarding sensitive customer data and ensuring privacy.
- **Intellectual Property:** PHARMA's brand, including its name and logo, has been trademarked to prevent any potential legal disputes. The system architecture and unique software features are also being considered for patenting to protect the company's intellectual property.

5. Risk Feasibility

- **Risk Identification:** Key risks include potential cyber-attacks, changes in pharmaceutical regulations, and supply chain disruptions. There is also the risk of strong competition driving down prices.
- **Mitigation Strategies:** To mitigate these risks, PHARMA will implement advanced cybersecurity measures, including regular penetration testing and encryption of all sensitive data. The legal team will keep abreast of regulatory changes, ensuring that the business remains compliant. Additionally, PHARMA will diversify its supplier base to reduce dependency on any single source, thus minimizing the impact of potential disruptions.

8. Project Planning

Project planning is concerned with identifying the following for every project:

- Activities
- Milestones
- Deliverables.

A plan must be drawn up to guide the development towards the project goal. A plan is drawn up at the start of a project. This plan should be used as the driver for the project. The initial plan is not static, and must be modified as the project progresses. Planning is required for development activities from specification through to delivery of the system.

9. Project Scheduling

GANTT chart

Task	Person(s) Responsible	Week 1	Week 2	Week 3	Week 4	Week 5	Week 6
Communication							
Quick Plan							
Modeling Quick Design							

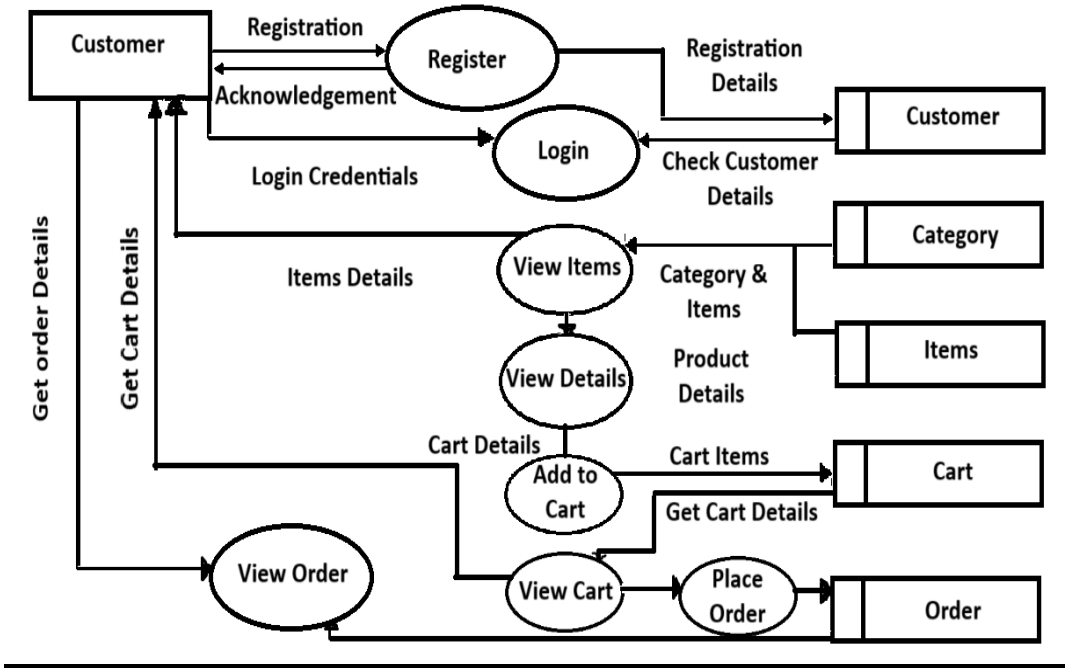
Construction of Prototype							
Deployment, Delivery and Feedback							

10. Software Engineering Paradigm Applied

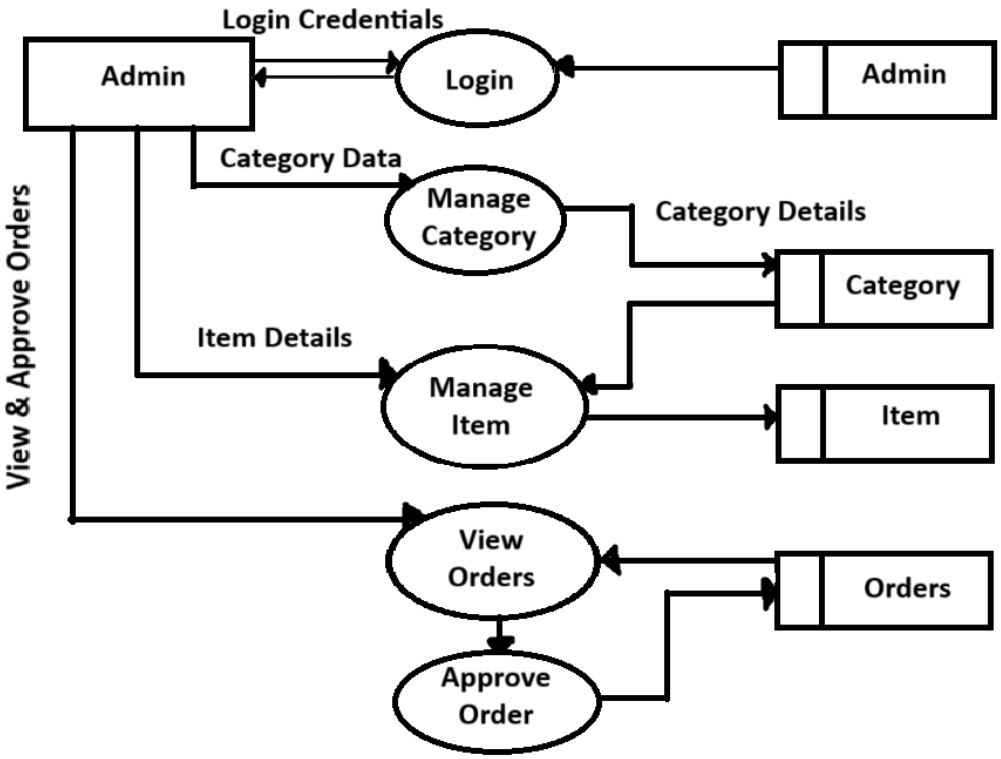
- **Level-0**



- **Level-1**



- **Admin Level-1**



11.Schema/Database Design (project)

Table Name admin:-

Name of Type Attribute	Data Type	Key	Description
id	int (11)	Primary key	Admin id
action_time	datetime (6)		
object_id	longtext		
object_repr	varchar(200)		
action_flag	smallint(5)		

change_message	longtext		
content_type_id	int(11)	Foreign key	
user_id	bigint(20)	Foreign key	

Table Name category:-

Name of Type Attribute	Data Type	Key	Description
cat_id	int(11)	Primary key	Category id
cat_name	varchar(150)		Category name

Table Name product:-

Name of Type Attribute	Data Type	Key	Description
p_id	int(11)	Primary key	Product id
p_name	varchar(150)		Product name
p_description	longtext		Product description
p_price	decimal(10,2)		Product price
p_image	varchar(100)		Product Image
category_id	int(11)	Foreign key	Category id

Table name s_product:-

Name of Type Attribute	Data Type	Key	Description
s_id	int(11)	Primary key	Product id
s_name	varchar(150)		Product name
s_description	longtext		Product description
s_price	decimal(10,2)		Product price
s_image	varchar(100)		Product image
category_id	int(11)	Foreign key	Category id

Table Name customUser:-

Name of Type Attribute	Data Type	Key	Description
id	Bigint(20)	Primary key	User id
password	varchar(128)		User password
last_login	datetime(6)		User last login
is_superuser	tinyint(1)		Is user is superuser
username	varchar(150)		User username
first_name	varchar(150)		User first name
last_name	varchar(150)		User last name
email	varchar(254)		User email
is_staff	tinyint(1)		Is user is staff
is_active	tinyint(1)		Is user is active
date_joined	datetime(6)		User join date

mobile	varchar(12)		User mobile number

Table Name contact:-

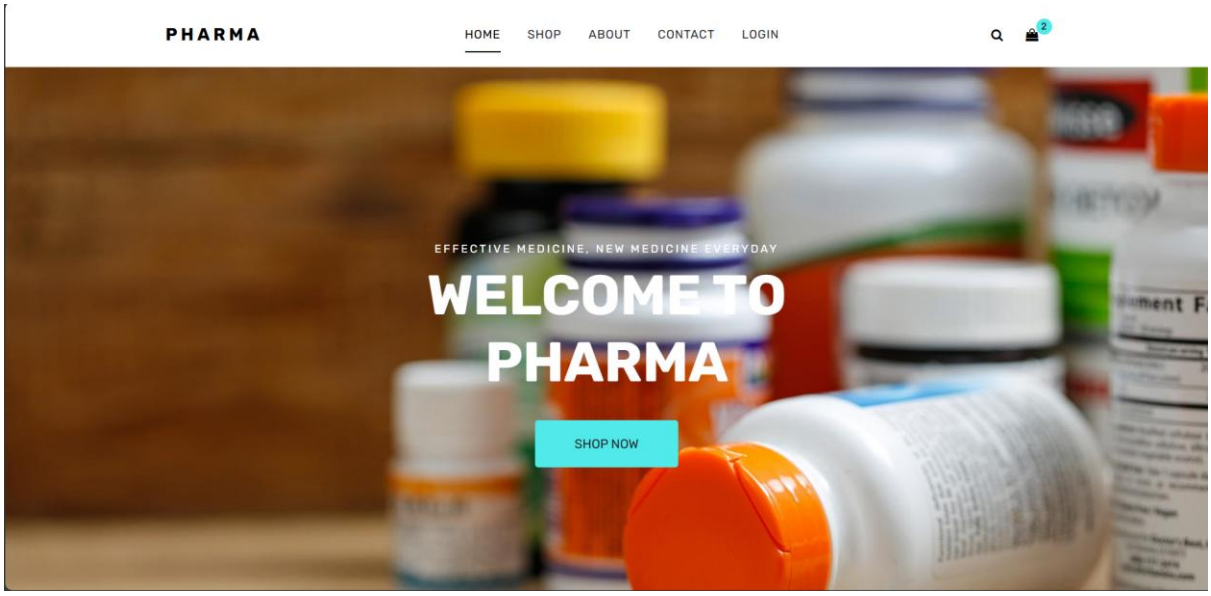
Name of Type Attribute	Data Type	Key	Description
id	bigint(20)	Primary key	id
first_Name	varchar(250)		First name
last_Name	varchar(250)		Last name
email	varchar(254)		email
subject	varchar(250)		Subject
message	longtext		Message

Table Name order:-

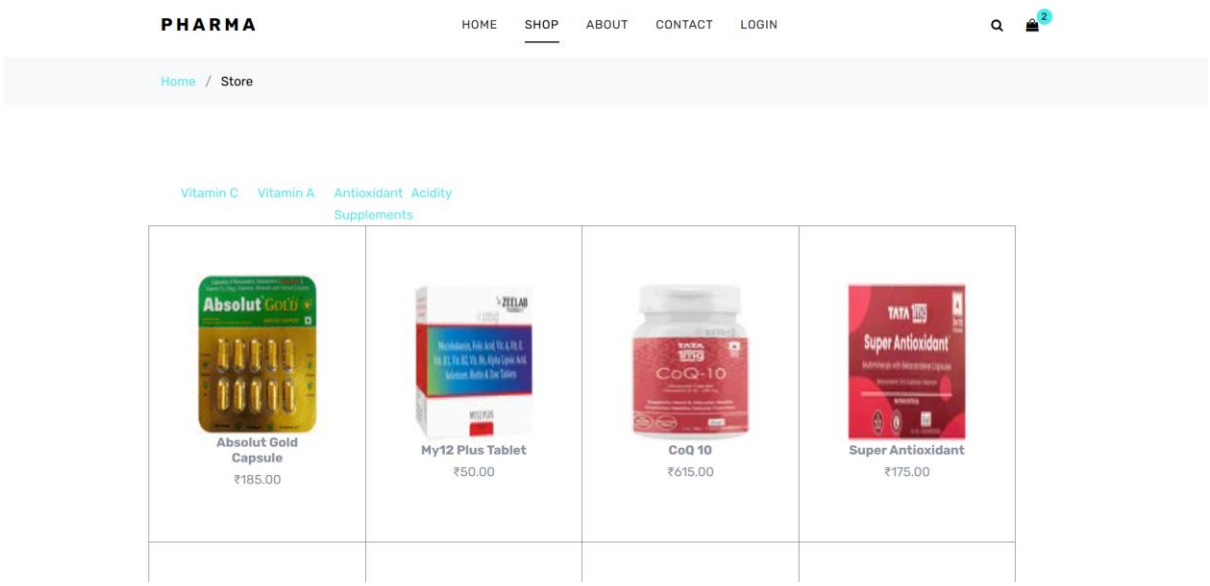
Name of Type Attribute	Data Type	Key	Description
id	bigint(20)	Primary key	Order id
quantity	int(10)		Order quantity
date_ordered	datetime(6)		Order date
payment_status	varchar(255)		Payment status
payment_id	varchar(255)		Payment id
address	longtext		Address
product_id	int(11)	Foreign key	Product id
user_id	bigint(20)	Foreign key	User id

User Interface Design

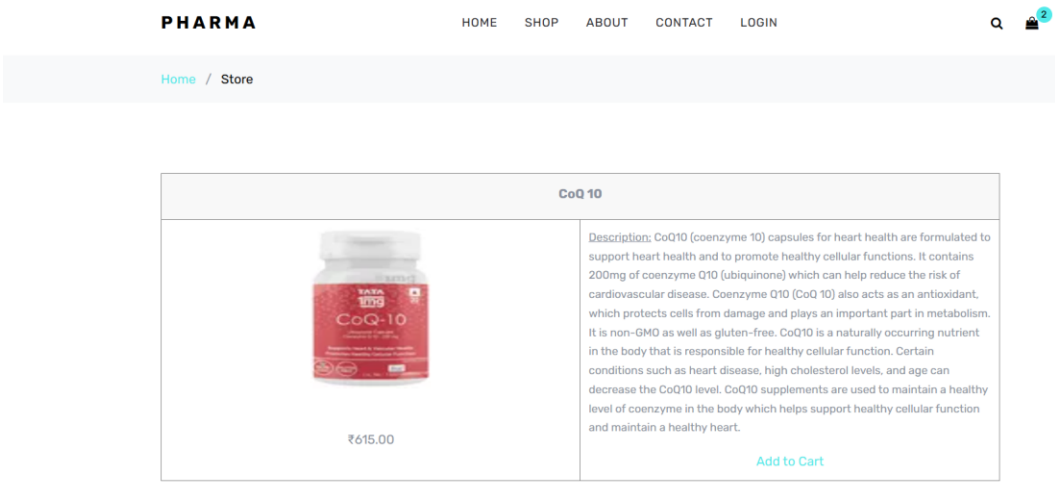
Home Page



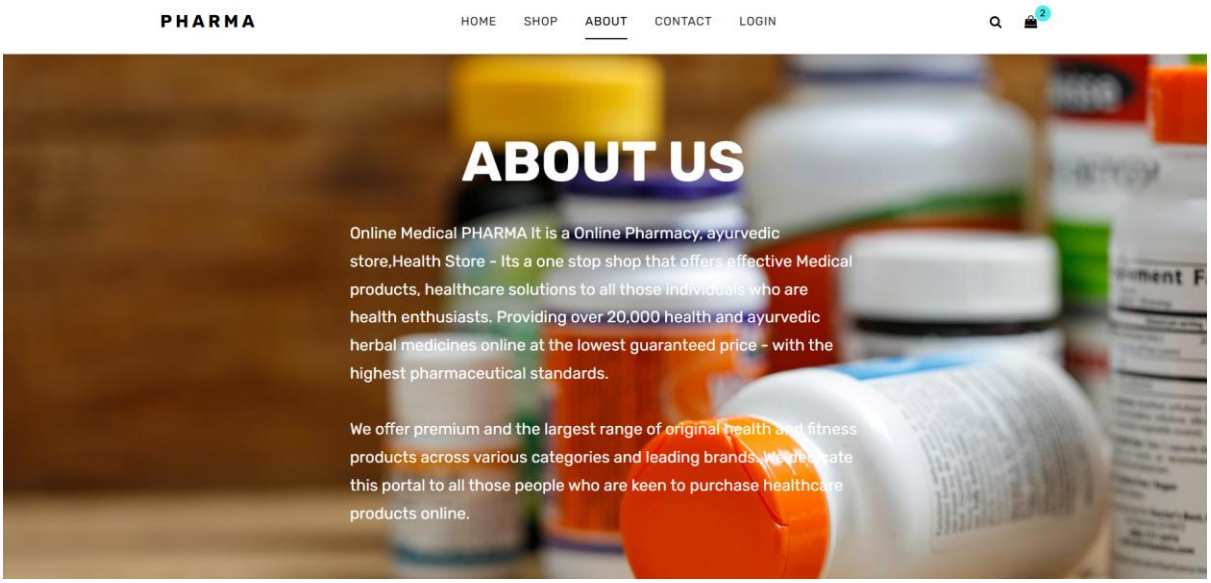
Shop Page



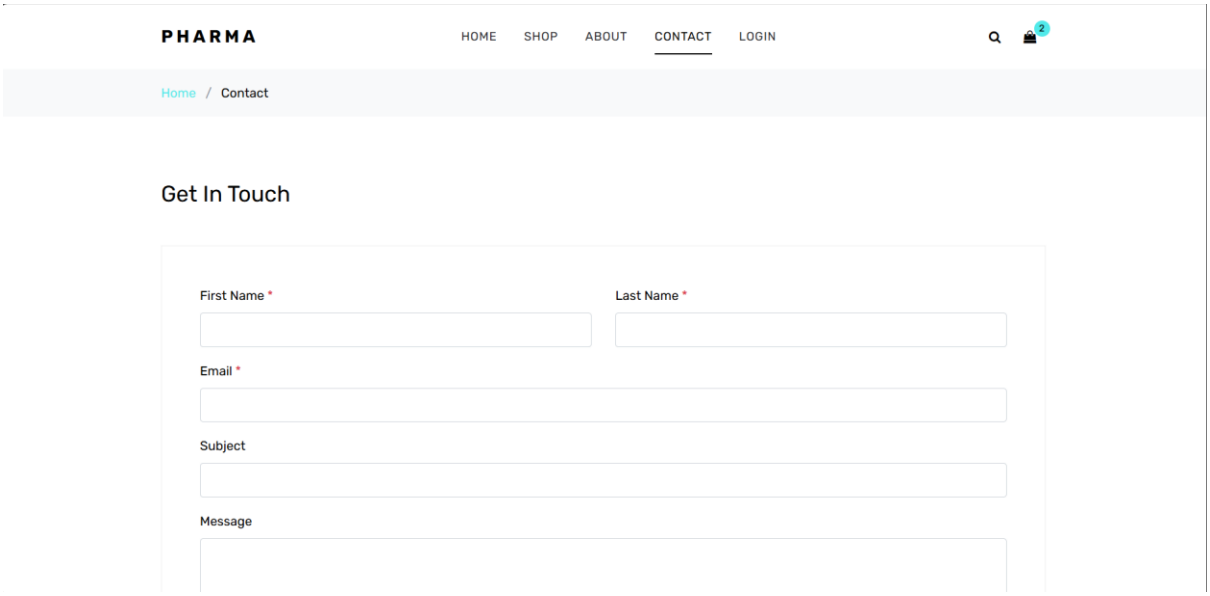
Product Page



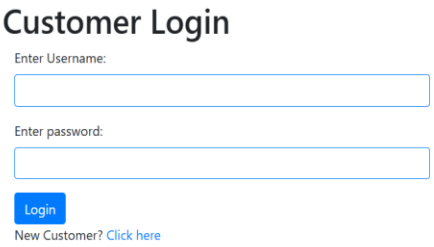
About Page



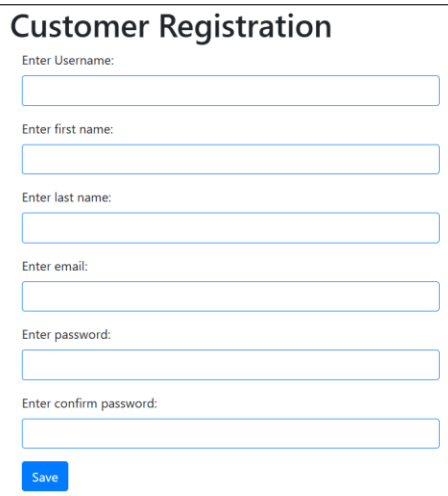
Contact Page



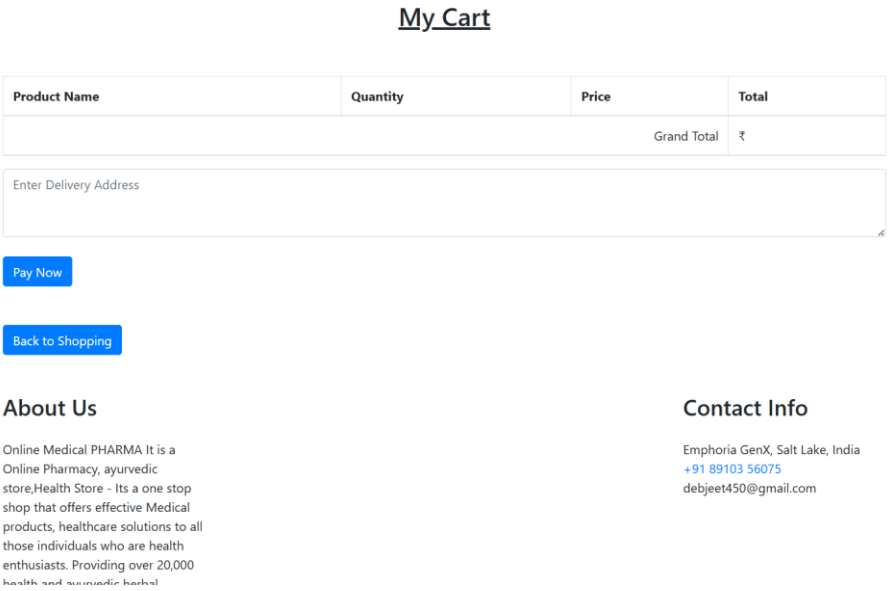
Login page



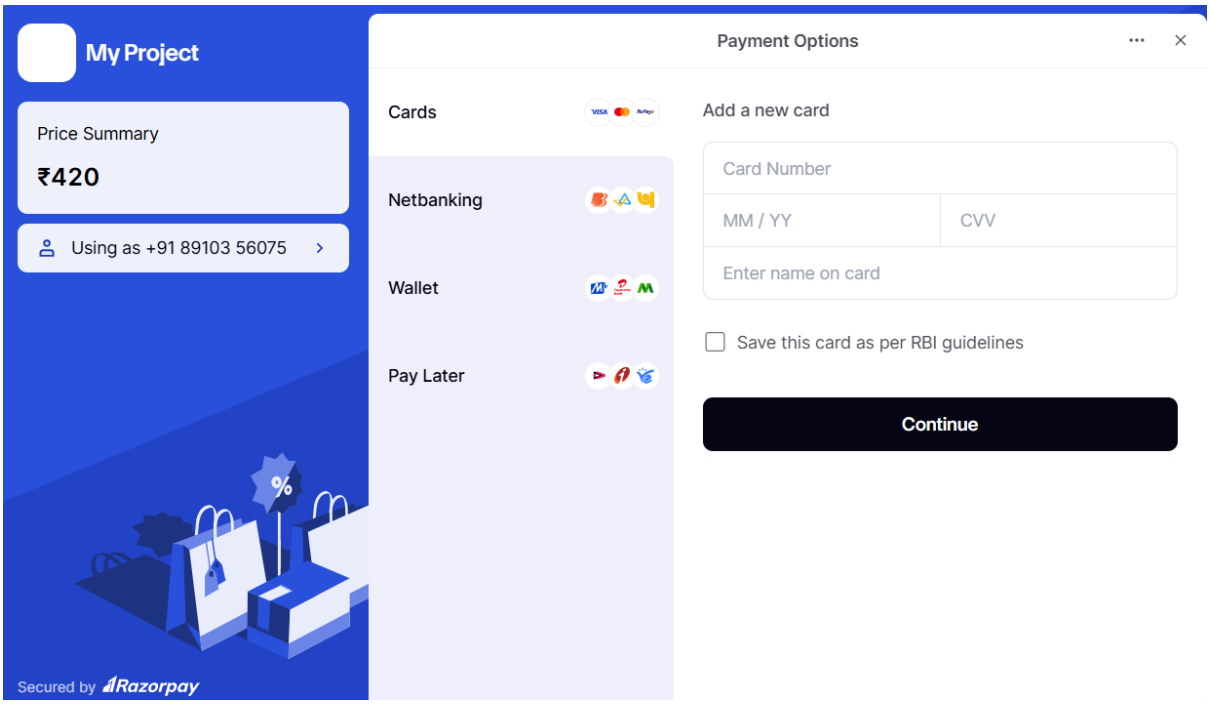
Registration page



Cart Page



Payment Page



12.Testing

1. Team Interaction

- Cross-Functional Collaboration: The testing team will work closely with developers, business analysts, and UX/UI designers to ensure that all requirements are well-understood and met. Regular stand-up meetings will be held to discuss progress, challenges, and priorities.
- Feedback Loop: The QA team will provide continuous feedback to the development team during the testing phases. Any identified bugs or issues will be communicated in real-time, allowing for quick fixes and retesting.
- Stakeholder Communication: Test results and progress updates will be regularly communicated to stakeholders, ensuring transparency and enabling informed decision-making regarding feature releases and project timelines.

2. Test Objective

- **Goal:** To ensure that the PHARMA online medicine shop is fully functional, secure, user-friendly, and capable of handling expected loads before its official launch. The testing process aims to identify and resolve any defects, ensuring a smooth and reliable user experience.

3. Process Overview

1. **Requirement Analysis:** Gather and analyse the functional and non-functional requirements to ensure that the test cases cover all scenarios.
2. **Test Planning:** Develop a comprehensive test plan that outlines the scope, objectives, resources, schedule, and deliverables of the testing process.
3. **Test Environment Setup:** Configure the testing environment, including necessary hardware, software, network configurations, and data sets to mimic the production environment.
4. **Test Case Development:** Create detailed test cases that cover all possible user interactions, including edge cases and negative scenarios.
5. **Test Execution:** Execute the test cases in a systematic manner, starting with unit tests, followed by integration, system, and user acceptance tests.
6. **Defect Reporting and Tracking:** Log any defects found during testing in a defect tracking system like JIRA, assigning priorities and ensuring that they are resolved promptly.
7. **Regression Testing:** Re-test the application after fixes and new feature additions to ensure that no new issues are introduced.
8. **Performance Testing:** Conduct load and stress tests to ensure that the system can handle high traffic and transaction volumes.
9. **Security Testing:** Perform penetration tests and security scans to identify vulnerabilities and ensure the safety of user data.

4. Testing Process (with Diagram)

Topics Discussed in the Diagram:

- **Requirement Analysis:** The initial phase where the requirements are reviewed to create relevant test cases.
- **Test Planning:** Outlines the strategy, resources, and schedule for the testing phases.
- **Test Case Development:** Specific scenarios are documented to cover all user interactions with the system.
- **Test Execution:** The actual running of test cases, checking for defects, and verifying that the application meets the required standards.
- **Defect Reporting and Tracking:** Issues found during testing are logged, tracked, and addressed by the development team.
- **Regression Testing:** Ensures that recent changes haven't negatively impacted the existing functionality.
- **Performance Testing:** Evaluates how the system performs under expected and peak load conditions.

- **Security Testing:** Focuses on identifying and mitigating security risks, ensuring data protection and compliance with legal standards.
- **Final Review:** Before launch, a final review of the testing outcomes ensures that all major issues are resolved.

5. Testing Strategy

- **Automated Testing**: Use automated testing tools for repetitive tasks such as regression testing, performance testing, and security scans. This will help in efficiently covering a wide range of test cases and scenarios.
- **Manual Testing:** Conduct manual testing for user interface evaluations, exploratory testing, and cases that require human judgment, such as user experience assessments.
- **Agile Approach:** Adopt an agile testing methodology, where testing is integrated into the development process, allowing for continuous feedback and iterative improvements.
- **Risk-Based Testing**: Prioritize testing efforts on areas with the highest risk, such as payment processing, user authentication, and data protection, to ensure critical functions are secure and reliable.

6. Imaginary Test Cases

- **Test Case 1: User Registration**
  - **Objective:** Verify that a new user can successfully register on the PHARMA platform.
  - **Steps:**
    1. **Navigate to the registration page.**
    2. **Enter valid user details (name, email, password, etc.).**
    3. **Submit the registration form.**
  - **Expected Result:** The user is successfully registered, receives a confirmation email, and can log in with the new credentials.
- Test Case 2: Add to Cart Functionality
  - **Objective**: Ensure that the "Add to Cart" functionality works correctly.
  - **Steps:**
    1. **Search for a medicine on the platform.**
    2. **Select the desired product from the search results.**
    3. **Click on "Add to Cart".**
    4. **Navigate to the cart to review the added item.**

  - **Expected Result:** The selected item appears in the cart with the correct quantity and price, and the user can proceed to checkout.

Unit Testing:

- **Objective:** Focuses on testing individual components or functions within the PHARMA application, such as the add_to_cart() function or user_login() method.
- **Process**: Each function or method is tested in isolation with different inputs to ensure it returns the expected outputs. For example, testing the calculate_total_price() function ensures that it accurately sums the prices of items in the cart.

White Box Testing:

- **Objective:** Involves testing the internal logic and structure of the code in PHARMA, ensuring that all paths, branches, and conditions execute as intended.
- **Process:** The tester, with knowledge of the codebase, examines the logic flow, such as verifying that the if-else conditions in the payment processing module correctly handle different payment methods.

Black Box Testing:

- **Objective:** Focuses on testing the functionality of PHARMA without considering the internal code structure. The tester only knows the inputs and expected outputs.
- **Process:** Test scenarios like searching for a medicine and verifying that the correct results are displayed, or submitting a registration form and checking for successful account creation are carried out without examining the underlying code.

System Testing:

- **Objective:** Ensures that the entire PHARMA system works together as a whole and meets the specified requirements.
- **Process:** Comprehensive testing is conducted on the full application, including user registration, product search, order placement, and payment processing, simulating real-world usage to ensure all components function correctly when integrated.

13.System Security measures (Implementation of security for the project developed)

- Only authorized users are allowed.

- Without signing in users are not allowed to go an intermediate page by typing an URL. For all such efforts, users will be redirected to the home page.

14.Database/Data security

- Database is present in remote machine.
- Oracle's default securities are applied.

15.Creation of User profiles and access rights

- The admin must create users manually
- The admin can create more admins

16.Future scope and further enhancement of the Project

Online medicine shop PHARMA can expand to offer personalized health recommendations, virtual consultations with doctors, and AI-powered disease diagnosis. Integrating with wearable health devices can enable real-time monitoring and medication reminders. Blockchain technology can ensure secure and transparent drug supply chains. Additionally, exploring drone delivery for remote areas can improve accessibility.

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15. **Food and Drug Administration (FDA):** https://www.fda.gov/
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17. **National Institutes of Health (NIH):** https://www.nih.gov/